A Closed-Loop Real-Time EEG Framework with Adaptive Denoising and Online Learning for Emotion Classification

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Abstract

We present a Python-based real-time EEG processing framework for stress analysis, evaluated on the Laparoscopic Surgery EEG dataset with continuous stress annotations. The pipeline, inspired by the REST toolbox, integrates band-pass filtering, Adaptive Subspace Reconstruction (ASR), and Online Recursive Independent Component Analysis (ORICA) with ICLabel-based artifact identification to achieve adaptive and automated artifact removal. Unlike offline ICA methods that yield a static global solution, ORICA dynamically tracks nonstationary brain sources, enabling effective artifact suppression and improved signal stability. Experimental results demonstrate that the framework effectively removes ocular and muscular artifacts and preserves task related neural activity, providing a robust foundation for real-time stress monitoring in realistic environments.

Introduction

- ***Motivation.*** EEG-based stress analysis plays an important role in healthcare, mental health monitoring, and cognitive workload assessment. Real-time EEG processing enables continuous monitoring and adaptive feedback, yet achieving robust analysis in real-world settings remains challenging due to ocular and muscular artifacts and the nonstationary nature of brain activity. Traditional offline ICA methods rely on static global decomposition and cannot adapt to dynamic signal changes, limiting their use in real-time EEG tasks.
- ***Our Work.*** To address these limitations, we implemented a standalone real-time EEG processing framework in Python. The framework is inspired by the Real-time EEG Source-mapping Toolbox (REST) and has been validated to maintain algorithmic consistency with REST. We further extended it by integrating ICLabel's artifact classification into an online module, allowing adaptive source cleaning and real-time feedback.
- ***Contribution.***
- *(1)* Development of a REST-consistent real-time EEG adaptive denoising framework based on ORICA and ASR in Python.
- *(2)* Validation of algorithmic consistency with REST and implementation of online ICLabel-based artifact classification.
- *(3)* Demonstration of scalability and deep-learning compatibility, providing a solid foundation for real-time stress and emotion decoding.

EEG Adaptive Denoising Pipeline

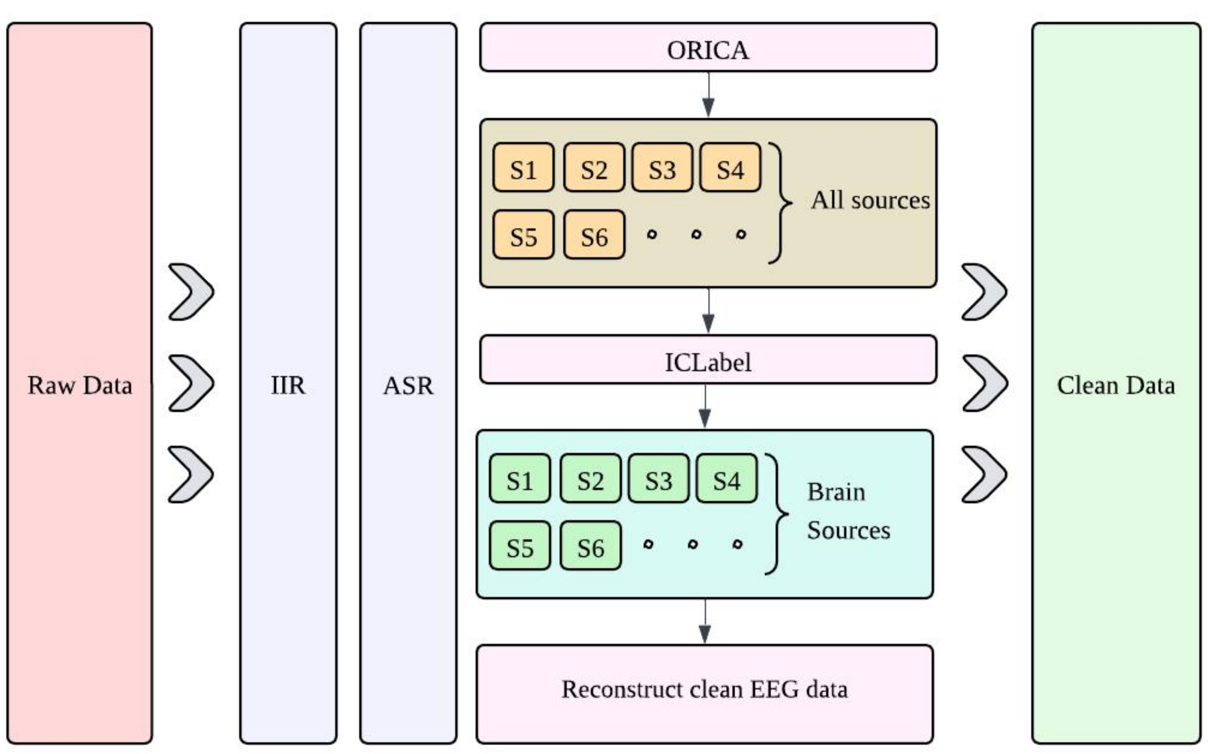

Figure 1. EEG Adaptive Denoising Framework

Components description:

- ✓ **IIR filtering:** removes baseline drift and high-frequency noise, with user-adjustable frequency ranges available in the GUI.
- ✓ **ASR:** suppresses transient artifacts.
- ✓ **ORICA:** performs adaptive source separation in real time.
- ✓ **ICLabel:** automatically classify and remove artifact-related sources.

System Implementation and Visualization

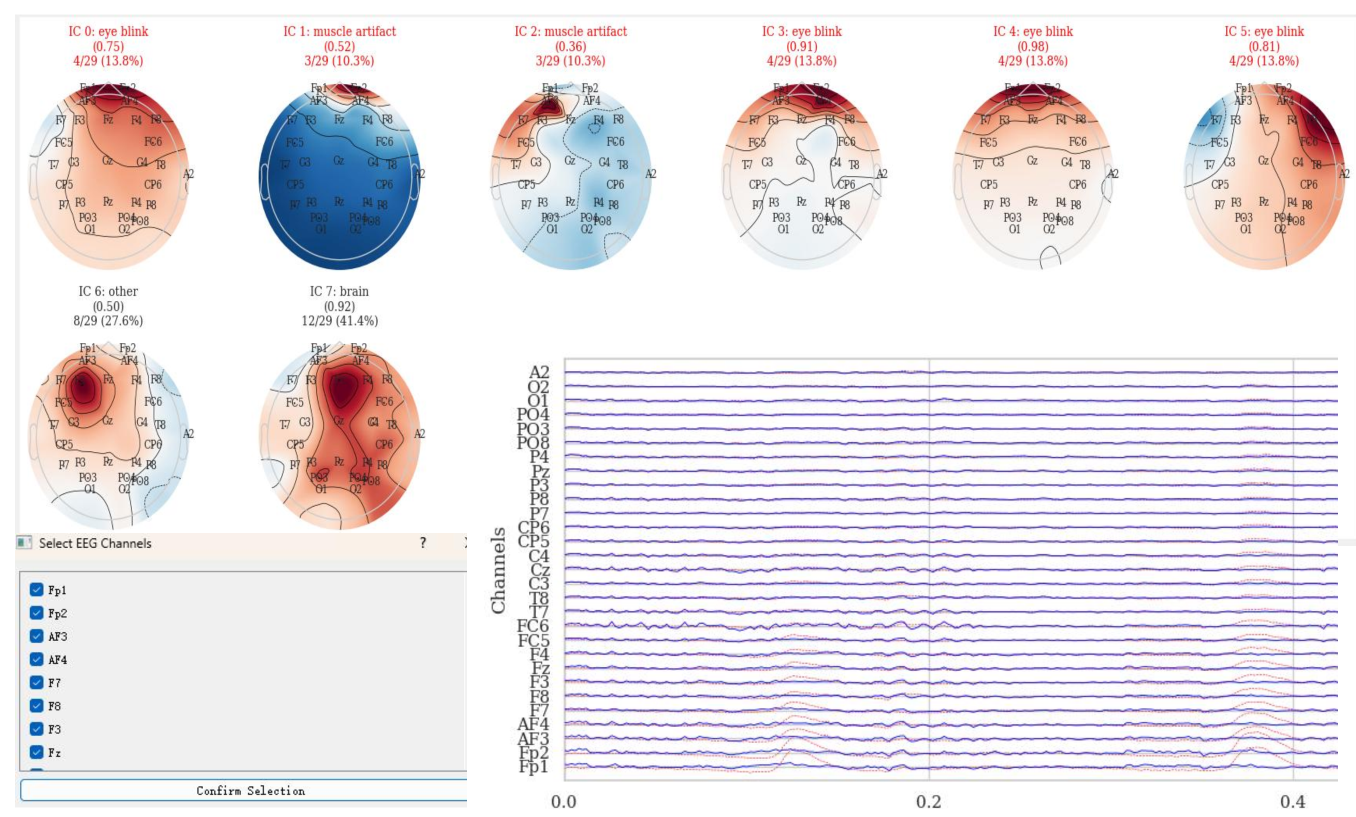

Figure 2. Visualization of the Python-based EEG Denoising System

Real-time Stress Classification

- ***Dataset.*** Laparoscopic Surgery Training EEG with continuous stress labels.
- ***Feature Selection.*** magnitude-squared coherence, bandpower. Both of them can be efficiently computed in real time.

$$\mathrm{BandPower}_x(f_1, f_2) = \frac{1}{N_f} \sum_{f_i \in [f_1, f_2]} P_{xx}(f_i)$$

$$COH_{xy} = \frac{|S_{xy}|^2}{S_{xx} S_{yy}}$$

- ***Offline Calibration.*** Pretrained SVM and XGBoost models were used as baseline classifiers for stress classification, while both ASR and ORICA modules were initialized to accelerate the convergence process.
- ***Online Testing.*** EEG data with continuous stress labels from the dataset were used to simulate real-time input and evaluate online stress recognition performance, comparing accuracy with and without artifact removal.

Future Work

- During the experiments, we observed a correlation between the instantaneous changes in stress levels and the mutual information (MI) of the sources separated by ORICA. This finding may further reveal deeper neural mechanisms associated with stress.
- If the simulated testing yields satisfactory results, we plan to apply the entire pipeline in real-world scenarios to achieve real-time stress classification.

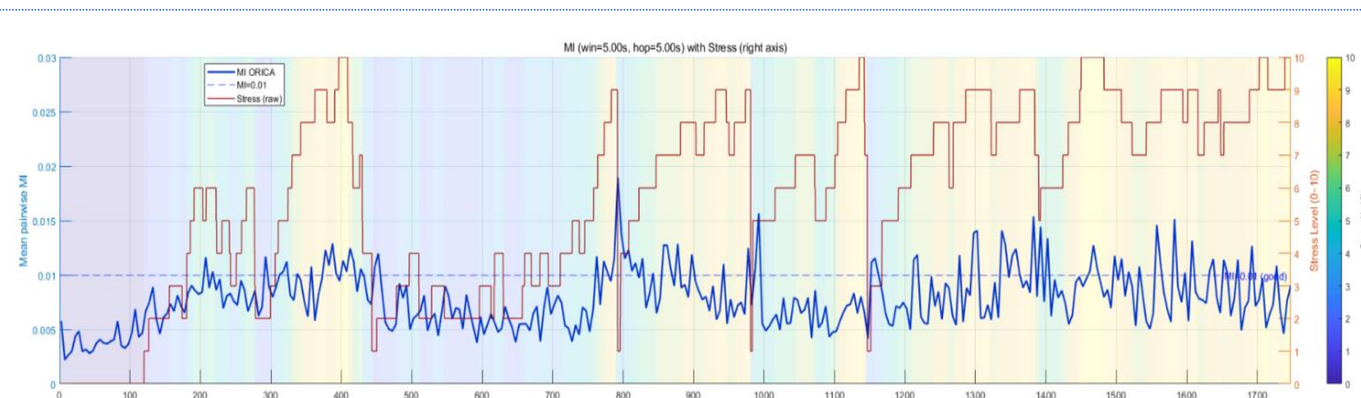

Figure 3. Correlation between Stress Level and ORICA-Derived Source Mutual Information